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ECG Circuit Design & Measuring

Abstract

The aim of this laboratory was to design, construct, and evaluate a complete electrocardiogram (ECG) signal acquisition system. The system designed consisted of a multi-stage signal conditioning circuit combining an instrumentation amplifier and an active band-pass filter, with an objective to amplify weak bioelectric signals while rejecting noise and interference. The circuit was aiming to reach an overall gain of approximately 625 V/V with a passband of 0.5 Hz to 100 Hz. The circuit was validated experimentally using both test signals as well as real ECG recordings, where the results demonstrated successful acquisition of ECG signals, while also highlighting the effects of component tolerances, mains interference, and motion artefacts on ECG signal quality.

Applied Theory

Electrocardiogram (ECG)

The ECG measures the electrical activity of the heart using electrodes placed on the body, at the Right Arm (RA), Right Leg (RL) and Left Leg (LL) electrode sites. The resulting signal reflects the depolarisation and repolarisation of cardiac tissue, producing the characteristic P wave, QRS complex, and T wave. These signals are typically very small (1–5 mV) and are often corrupted by noise, hence the required amplification and filtering.

Instrumentation Amplifier

The instrumentation amplifier amplifies the differential input voltage while rejecting common-mode signals. Its output is given by: $V_o = A_d(V_A - V_B)$. The gain is determined by resistor ratios across the amplifier stages and a high common-mode rejection is essential in order to try to minimise interference such as 50 Hz mains noise.

Band-Pass Filtering

The band-pass filter is implemented by cascading a high-pass and low-pass filter:

$$f_L = \frac{1}{2\pi RC}, \quad f_H = \frac{1}{2\pi RC}$$

The circuit seen in the schematic below in Figure.1 was designed for a passband of 0.5–100 Hz, matching the frequency range of ECG signals, and was designed with the intent of reaching an ideal, theoretical total gain of: $A_{total} = A_1 \times A_2 = 25 \times 25 = 625 \text{ V/V}$

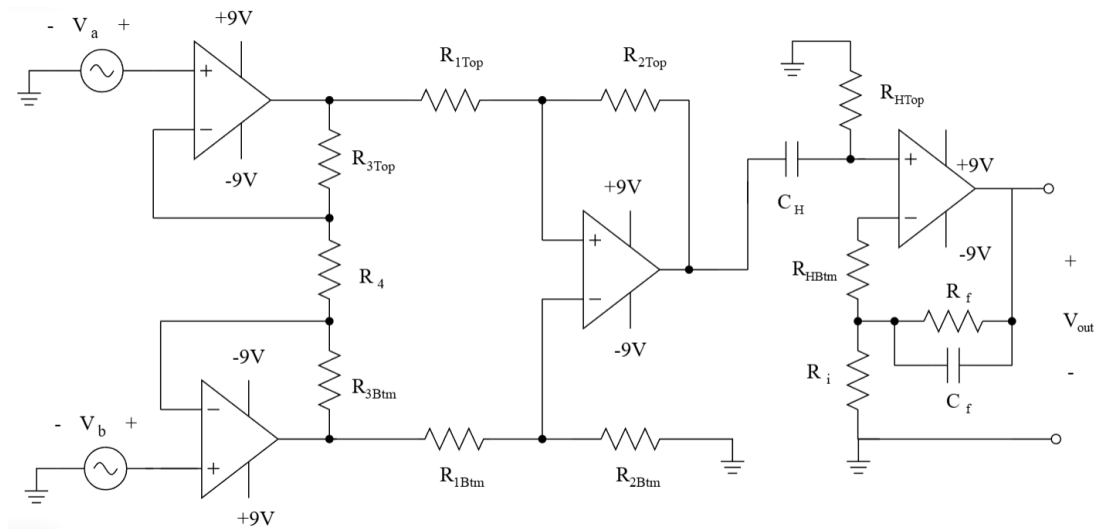


Figure.1: ECG circuit schematic

Experimental Method

Component Values and Actual Circuit Calculations

Component	Ideal Value	Measured Value
R_{1_Top}	10k Ω	9.64k Ω
R_{1_Btm}	10k Ω	9.86k Ω

R _{2_Top}	47kΩ	46.23kΩ
R _{2_Btm}	47kΩ	46.82kΩ
R _{3_Top}	18kΩ	17.79kΩ
R _{3_Btm}	18kΩ	17.78kΩ
R ₄	8.2kΩ	8.23kΩ
R _{H_Top}	330kΩ	328.3kΩ
R _{H_Btm}	330kΩ	329kΩ
R _i	56Ω	55.8Ω
R _f	1.5kΩ	1.48kΩ
C _H	1μF	0.991μF
C _f	1μF	0.995μF

Table.1: Circuit Component values (Nominal vs Measured)

Instrumentation Amplifier Gain

First stage gain:

$$A_{d1} = 1 + \frac{R_{3Top} + R_{3Btm}}{R_4} = 1 + \frac{17.79 + 17.78}{8.23} = 1 + 4.32 = 5.32 \text{ V/V}$$

The second stage gain is represented by:

$$A_{d2} = \frac{R_2}{R_1}$$

This ratio occurs on both a top branch:

$$A_{d2,Top} = \frac{46.23}{9.64} = 4.8 \text{ V/V}$$

As well as a bottom branch:

$$A_{d2,Btm} = \frac{46.82}{9.86} = 4.75 \text{ V/V}$$

Averaging across these values gives that $A_{d2} \approx 4.775 \text{ V/V}$

The theoretical value for the instrumentation amplifier gain is therefore:

$$A_{inst} = A_{d1} \times A_{d2} = 5.32 \times 4.775 = 25.38 \text{ V/V}$$

Active Filter Gain

$$A_{filter} = \frac{R_f}{R_i} = \frac{1.48}{0.0558} = 26.52 \text{ V/V}$$

Total Gain

$$A_{total} = 25.38 \times 26.52 = 673.2 \text{ V/V}$$

This is slightly higher than the theoretical value of 625 V/V, however this was somewhat inevitable given the non-ideal values available within the lab.

Bandpass filter design and Cutoff Frequencies

High-Pass Cutoff Frequency

The high-pass filter is formed by the coupling capacitor C_H and the resistor connected from the non-inverting input of the final amplifier stage to ground. This resistor provides a discharge path for the capacitor and sets the lower cutoff frequency of the circuit. This lower cutoff frequency is given by:

$$f_L = \frac{1}{2\pi R_H C_H}$$

So using the measured resistor and capacitor values $R_H = 328.3 \text{ k}\Omega$, $C_H = 0.991 \text{ }\mu\text{F}$:

$$f_L = \frac{1}{2\pi(328.3 \times 10^3)(0.991 \times 10^{-6})} = 0.489 \text{ Hz}$$

This value is very close to the intended lower cutoff frequency of approximately 0.5 Hz which is good. It should be noted that although a similar resistor is present on the inverting input of the op-amp, this is only included to balance input bias currents and does not actually contribute to the cutoff frequency calculation.

Low-Pass Cutoff Frequency

The low-pass filter is implemented as an active filter using the feedback resistor R_f and capacitor C_f . These components determine the upper cutoff frequency by limiting the gain at higher frequencies.

The cutoff frequency is given by a similar equation as before:

$$f_H = \frac{1}{2\pi R_f C_f}$$

Therefore with the matching measured values:

$$R_f = 1.48 \text{ k}\Omega, \quad C_f = 0.995 \text{ }\mu\text{F}$$
$$f_H = \frac{1}{2\pi(1.48 \times 10^3)(0.995 \times 10^{-6})} = 108.2 \text{ Hz}$$

Again this is quite close to the intended upper cutoff frequency of approximately 100 Hz.

Frequency Response Verification

Before measuring any ECG data the circuit was verified using test signals to check its operation was as intended. The system was tested at three frequencies of interest: 0.5 Hz, 1.1 Hz, and 100 Hz. These frequencies were selected as they correspond approximately to the lower cutoff frequency, a frequency within the passband, and the upper cutoff frequency respectively. The measured results from the Analogue Discovery DAQ unit were as follows in Table.2:

Frequency (Hz)	V _a (mV)	V _b (mV)	V _o (V)	Measured V _b -V _a (mV)	Gain (V/V)
0.5	5	10	2.46	4.8902	489.02
1.1	5	10	3.21	5.01	640.7
100	5	10	2.56	5.01	510.95

Table.2: DAQ measured system test values

At the cutoff frequencies of a first-order filter, the gain is expected to reduce to approximately $\frac{1}{\sqrt{2}} \approx 0.707$ of the mid-band gain, corresponding to the -3 dB point. Taking the measured mid-band gain from the DAQ of approximately 640 V/V, the expected gain at the cutoff frequencies is approximately 450 V/V. The experimentally measured gains at 0.5 Hz and 100 Hz were slightly higher than this value, which can likely be explained by the fact that these test frequencies are close to but not exactly equal to what the true cutoff frequencies should be when calculated from the component values. Additionally, any non-ideal effects such as component tolerances and op-amp limitations contribute to deviations from the ideal response. Regardless of this though it still demonstrates the intended features, with the bandpass filter working approximately as intended within its given passband/stopband regions.

ECG Recording Method

Having verified the circuit with the sinusoidal test signals, ECG recordings were taken from a subject using three disposable ECG electrodes. The electrodes were placed approximately in a Lead II configuration, with one electrode connected to the right arm/upper right chest, one to the left leg/lower abdomen, and one to the right leg/lower abdomen as the reference/ground electrode. The circuit was powered using two 9 V batteries during ECG recording, to ensure that the circuit was isolated from the mains supply for safety reasons. The subject was instructed to sit as still as possible while the ECG was recorded using the Analog Discovery device and WaveForms software.

Following this three separate 20-second recordings were taken:

1. A normal ECG recording while the subject remained still.
2. A recording while the subject held the insulation of a mains cable.
3. A recording while the electrode near the upper right shoulder/chest was mechanically disturbed.

Each recording was sampled at 400 Hz and exported to an Excel .csv file for analysis in MATLAB.

ECG Plotting and Results

The following MATLAB code was written in order to visualize the data from the ECG:

```
%Sampling freq from lab brief
fs = 400;
%Loading ECG data
normalData = readmatrix('DC_RW_norm.csv');
mainsData = readmatrix('DC_RW_50.csv');
ruffleData = readmatrix('DC_RW_ruffle.csv');
```

```
%Voltage signal from ECG
normalECG = normalData(:,end);
mainsECG = mainsData(:,end);
ruffleECG = ruffleData(:,end);
%Time vectors for each ECG
t_normal = (0:length(normalECG)-1)/fs;
t_mains = (0:length(mainsECG)-1)/fs;
t_ruffle = (0:length(ruffleECG)-1)/fs;
```

```

%Plotting normal ECG
figure;
plot(t_normal, normalECG);
xlabel('Time (s)');
ylabel('Voltage (V)');
title('Normal ECG Recording');
grid on;
%Plotting ECG with mains interference
figure;
plot(t_mains, mainsECG);
xlabel('Time (s)');
ylabel('Voltage (V)');
title('ECG Recording with 50 Hz Mains Interference');
grid on;
%Plotting ECG with electrode motion
figure;
plot(t_ruffle, ruffleECG);
xlabel('Time (s)');
ylabel('Voltage (V)');
title('ECG Recording with Electrode Motion Artefact');
grid on;

```

Code_Appendix.1: Plotting of raw ECG data

The output of this was the following three plots of Voltage over Time. Each of these ECG recordings seemingly demonstrated the expected behaviour of the signal under the different imposed conditions.

The normal ECG in Figure.2 shows a clear periodic waveform with identifiable QRS complexes, confirming that the circuit indeed successfully amplified the underlying cardiac signal. However, some baseline noise and a noticeable transient dip are present (at approximately 12-14 seconds into the recording), likely due to some minor electrode movement or baseline drift.

When the subject was exposed to mains interference, the ECG signal in Figure.3 becomes visibly contaminated with a higher-frequency oscillatory component upon the waveform, which should be due to the 50 Hz coupling from the surrounding electrical environment.

Lastly the recording in Figure.4, obtained during electrode motion, shows large, irregular fluctuations and significant baseline variation, which obscured the underlying ECG features and are in line with what should be characteristic of motion artefacts caused by changes in electrode–skin impedance.

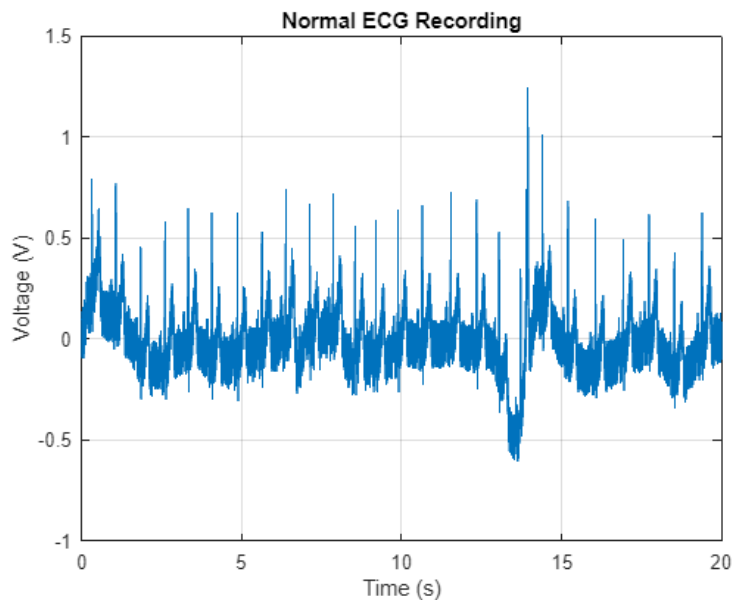


Figure.2: Normal ECG Recording

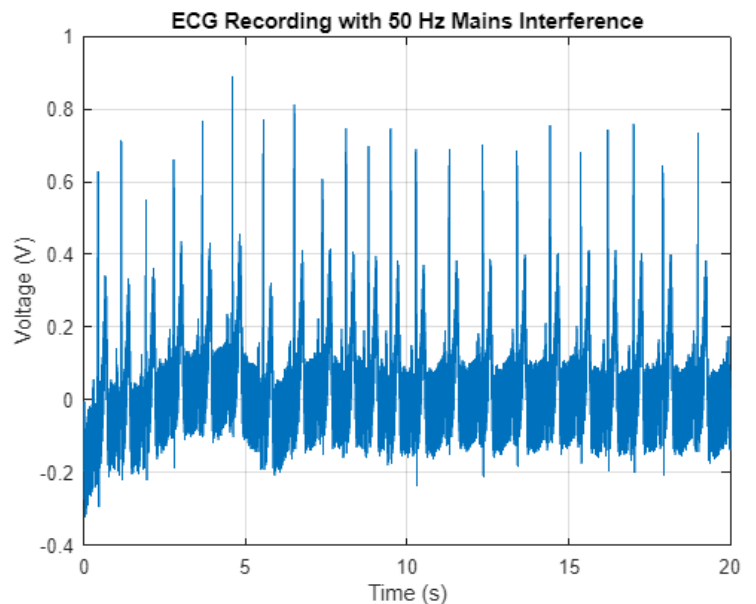


Figure.3: ECG Recording with 50Hz Mains Interference

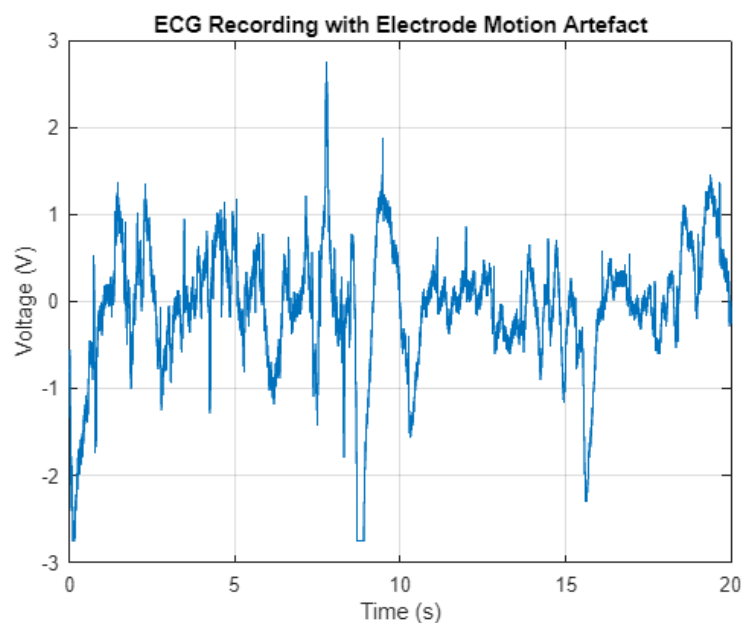


Figure.4: ECG Recording with Electrode Motion Artefact

To further confirm the characteristic features of the ECG signals and the effects of the imposed conditions, I used resources and prior knowledge from the EEEN30160 Biomedical Signal Processing ran by Elaine Corbett in order to convert this data into the Frequency domain plotting it against the Voltage magnitude in dB, which allowed for much better comparison of the 50Hz mains effect across the respective ECG plots. This also allowed for me to see where I should apply a notch filter in order to clean the raw Normal ECG data, which improved the clarity of the QRS complexes significantly when comparing the filtered and raw signals. The following MATLAB code function was written to implement these changes:

```
function [ecg_clean, freq_Hz, mag_raw_dB, mag_clean_dB] =
BIOINSTanalyseECGFrequency(filename, Fs, doClean, plotTitle)
%Inputs:
%filename = CSV filename, e.g. 'DC_RW_norm.csv'
%Fs = sampling frequency, for this lab use 400 Hz
%doClean = true or false
%plotTitle = title text for figures
```

```

%Reading in ECG data
data = readmatrix(filename);
ecg_raw = data(:,end);
ecg_raw = ecg_raw(:);
N = length(ecg_raw);
time_raw = (0:N-1)' / Fs;

%Plotting raw ECG in time domain
figure;
plot(time_raw, ecg_raw);
xlabel('Time (s)');
ylabel('Voltage (V)');
title(['Time Domain: ', plotTitle]);
grid on;

%Frequency spectrum of raw ECG
Nfft = 2^nextpow2(N);
X_raw = fft(ecg_raw, Nfft);
freq_Hz = (0:Nfft/2)' * (Fs/Nfft);
mag_raw = abs(X_raw(1:length(freq_Hz)));
%Normalising magnitude in dB
mag_raw_dB = 20*log10(mag_raw / max(mag_raw));

figure;
plot(freq_Hz, mag_raw_dB);
xlabel('Frequency (Hz)');
ylabel('Normalised Magnitude (dB)');
title(['Frequency Spectrum: ', plotTitle]);
xlim([0 120]);
ylim([-100 5]);
grid on;

%Cleaning of signal if wanted i.e for normal signal but not 50 Hz one
if doClean == true

    %Adding reflection padding to reduce edge effects
    padDuration_sec = 3;
    numPadSamples = min(round(padDuration_sec*Fs), floor(N/2)-1);
    prePadding = flipud(ecg_raw(1:numPadSamples));
    postPadding = flipud(ecg_raw(end-numPadSamples+1:end));

    ecg_extended = [prePadding; ecg_raw; postPadding];

    %50 Hz notch filter
    mainsFreq_Hz = 50;
    wo = mainsFreq_Hz / (Fs/2);
    Q = 55;
    bw = wo / Q;
    [notch_b, notch_a] = iirnotch(wo, bw);
    ecg_notched_ext = filtfilt(notch_b, notch_a, ecg_extended);

    %Removes padding
    ecg_clean = ecg_notched_ext(numPadSamples+1 : numPadSamples+N);

    %Plotting the cleaned ECG
    figure;
    plot(time_raw, ecg_clean, 'LineWidth', 1.1);
    xlabel('Time (s)');
    ylabel('Voltage (V)');
    title(['Cleaned ECG: ', plotTitle]);
    legend('Cleaned ECG');
    grid on;
    hold off;

```

```

%Plotting the raw vs cleaned ECG
figure;
plot(time_raw, ecg_raw);
hold on;
plot(time_raw, ecg_clean, 'LineWidth', 1.1);
xlabel('Time (s)');
ylabel('Voltage (V)');
title(['Raw vs Cleaned ECG: ', plotTitle]);
legend('Raw ECG', 'Cleaned ECG');
grid on;
hold off;

%Frequency spectrum of cleaned ECG
X_clean = fft(ecg_clean, Nfft);
mag_clean = abs(X_clean(1:length(freq_Hz)));
mag_clean_dB = 20*log10(mag_clean / max(mag_clean));

figure;
plot(freq_Hz, mag_raw_dB);
hold on;
plot(freq_Hz, mag_clean_dB, 'LineWidth', 1.1);
xlabel('Frequency (Hz)');
ylabel('Normalised Magnitude (dB)');
title(['Raw vs Cleaned Spectrum: ', plotTitle]);
legend('Raw ECG', 'Cleaned ECG');
xlim([0 120]);
ylim([-100 5]);
grid on;
hold off;

else
    ecg_clean = [];
    mag_clean_dB = [];
end
end

```

Code_Appendix.2: Function for plotting of processed ECG data

Figures.5-7 are each respective signals Frequency Spectra:

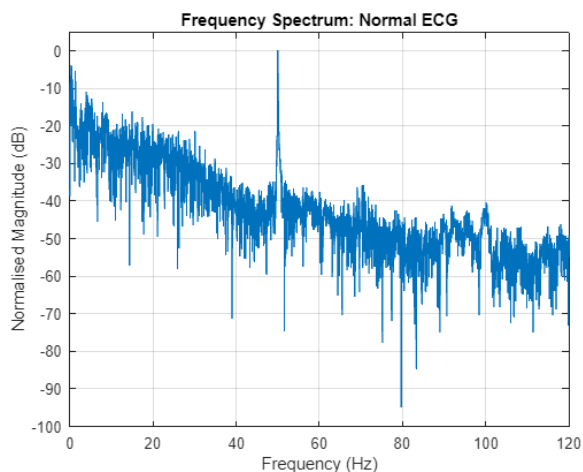


Figure.5: Normal ECG Frequency Spectrum

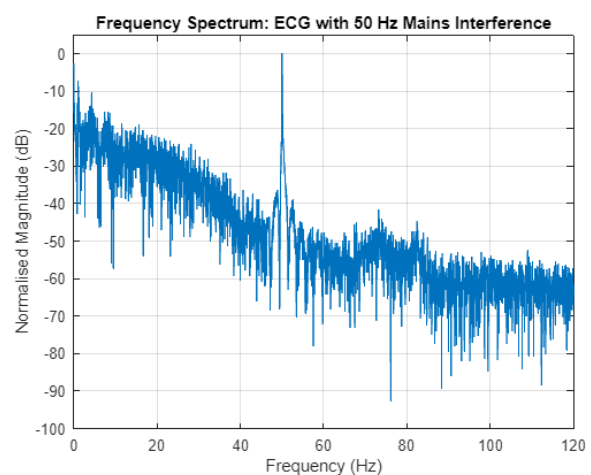


Figure.6: 50Hz Mains Interference ECG Frequency Spectrum

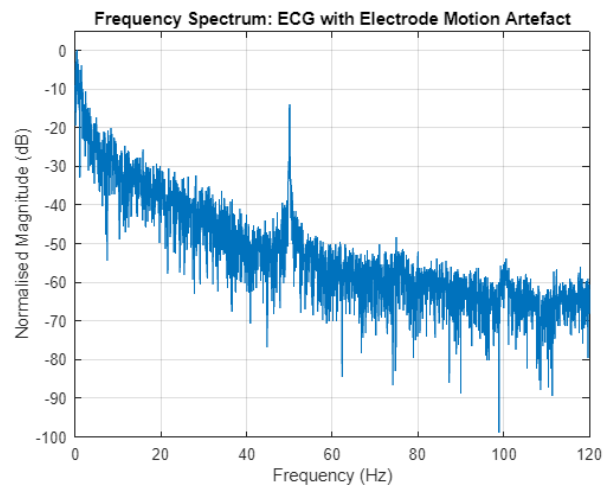


Figure.7: Motion Artefact ECG Frequency Spectrum

Across these spectra it is very evident that a 50Hz mains interference exists, most notably so in Figure.5 and Figure.6. This 50Hz noise has a slightly lower Magnitude in Figure.7, however this can likely be explained by the fact that the electrode motion introduces large-amplitude, broadband noise across a range of frequencies. As a result the overall spectral magnitude increases, reducing the relative prominence of the 50 Hz component when the spectrum is normalised. In contrast, when only the mains interference was present, the signal is otherwise relatively clean with the 50 Hz component appearing as a standalone sharp and dominant peak. Given that this 50Hz noise was very apparent across each recording I included an option of notch filtering it out for the likes of my normal ECG. This gave a much cleaner result as can be observed below in Figure.8 and Figure.9.

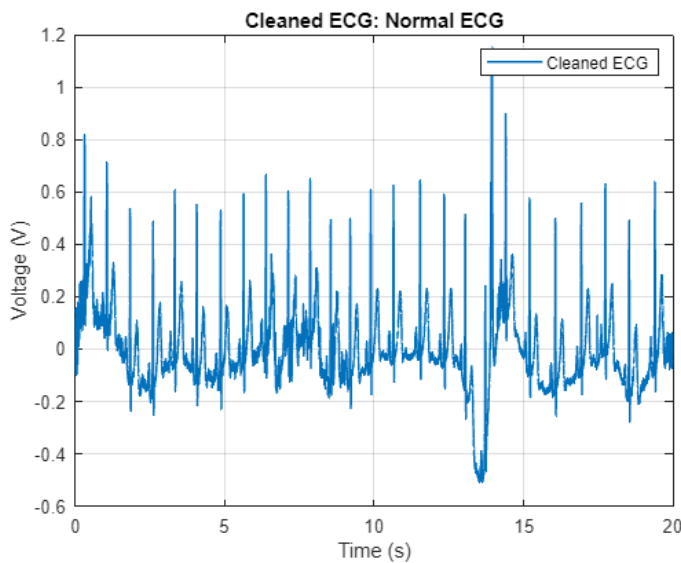


Figure.8: Normal ECG Clean of 50Hz noise

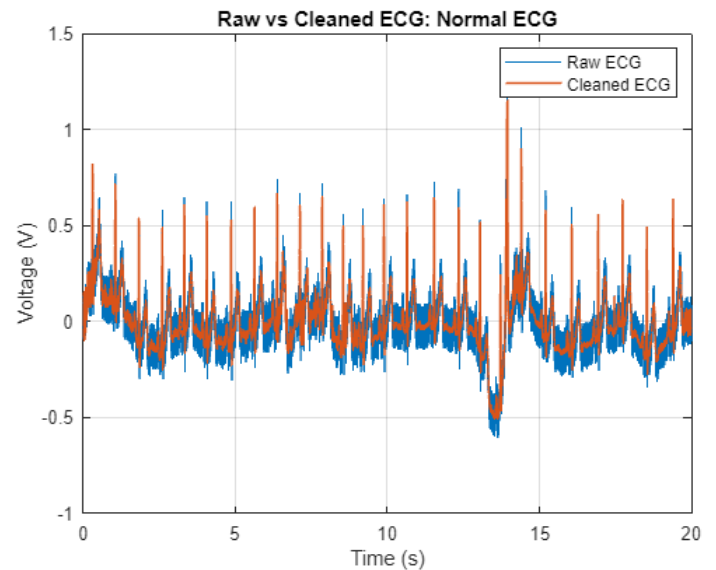


Figure.9: Raw vs Cleaned Normal ECG

Discussion

From the results across both the bench tests signals and the true electrode measured data, this ECG signal conditioning circuit was indeed successfully designed. It achieved an overall gain close to the intended value and a passband consistent with the required 0.5–100 Hz range. Frequency response testing confirmed that the circuit also exhibited the expected bandpass behaviour, with reduced gain near the cutoff frequencies and maximum amplification within the passband, with the QRS complexes being strongly visible in the notch filtered normal ECG when returned to the time domain. The real world disturbances including 50 Hz mains interference and motion artefacts were also clear within the signal, both of which were consistent with theoretical expectations. Overall the results highlight the effectiveness of the circuit in amplifying physiological signals while also illustrating the importance of noise mitigation and stable electrode placement in biomedical measurements given how sensitive the design was to these respective effects.